

Scalability Problems (Geographical)

Most based on Synchronous Communication.

Communication in WAN inherently unreliable.

Centralized Solutions.

Scalability Problems (Administratively)

How to scale a distributed system across multiple, independent administrative domains.

Major problem is conflicting policies with respect to:

- Resources usage.
- Management.
- Security.

Scaling Techniques (1)

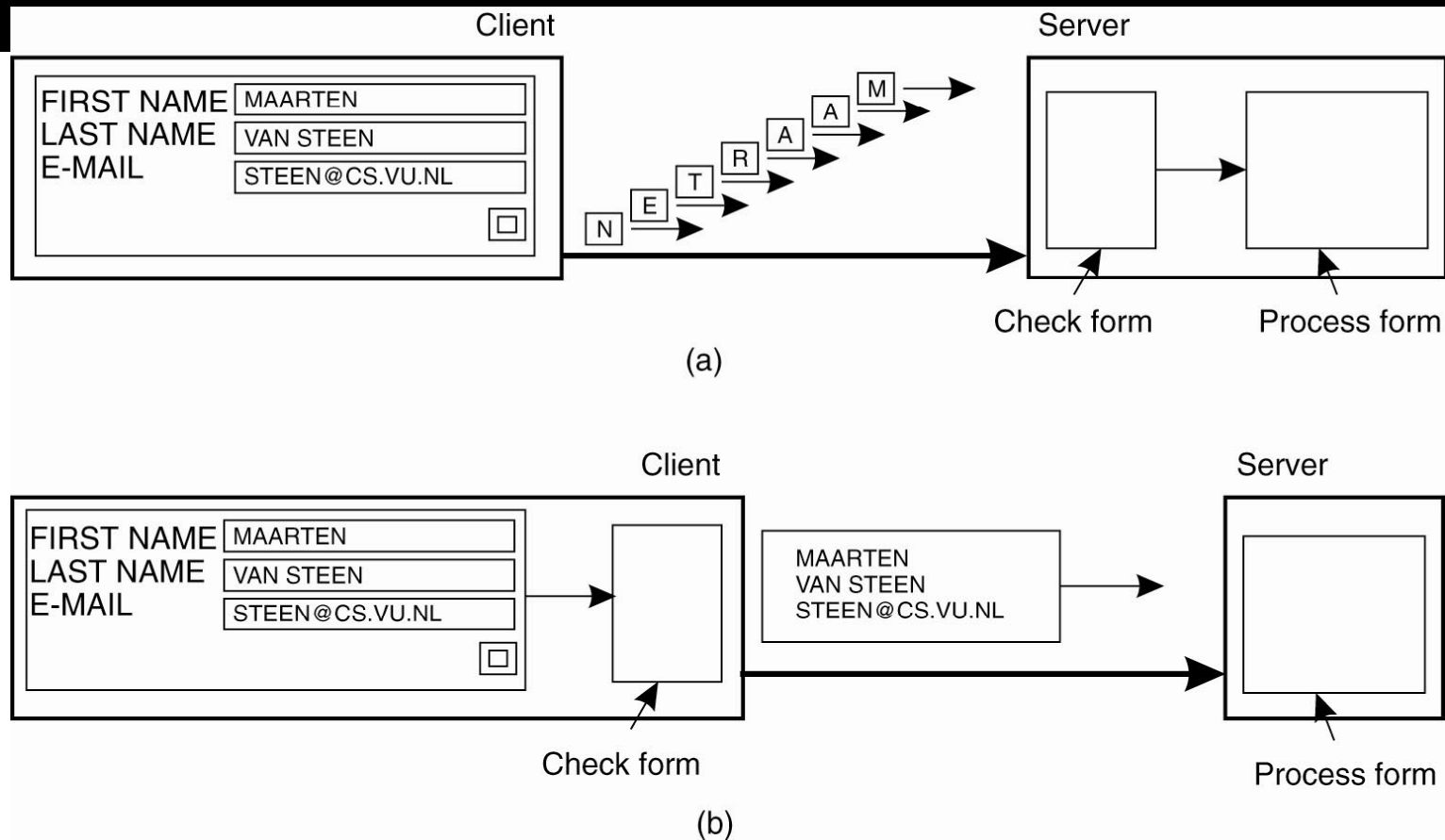


Figure 1-4. The difference between letting (a) a server or (b) a client check forms as they are being filled.

Scaling Techniques (2)

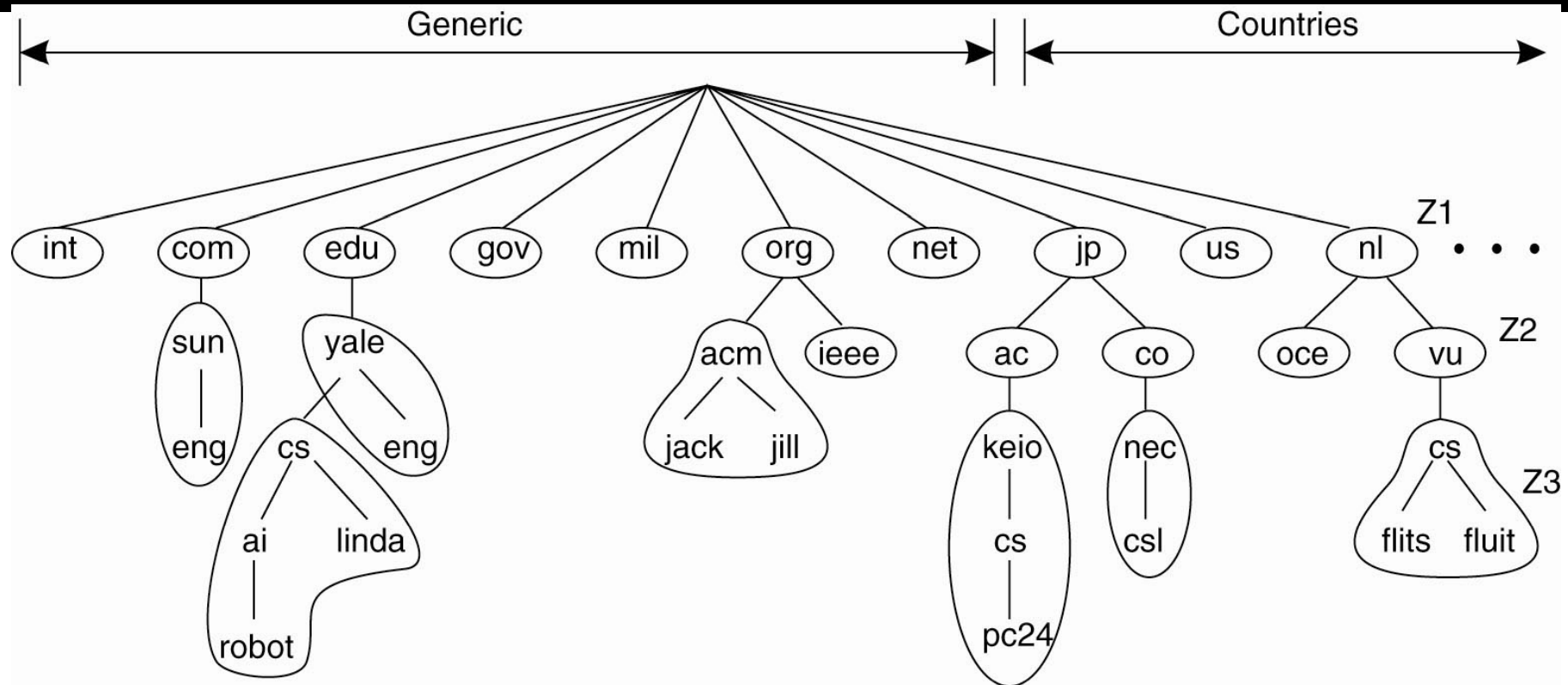


Figure 1-5. An example of dividing the DNS name space into zones.

Pitfalls when Developing Distributed Systems

False assumptions
made by first time
developer:

The network is
reliable.

The network is
secure.

The network is
homogeneous.

The topology does
not change.

Latency is zero.

Bandwidth is
infinite.

Transport cost is
zero.

There is one
administrator.

Types of Distributed Systems

Distributed Computing Systems.

Distributed Information Systems.

Distributed Pervasive Systems.

Two major Subgroups of Distributed Computing Systems.

Cluster Computing:

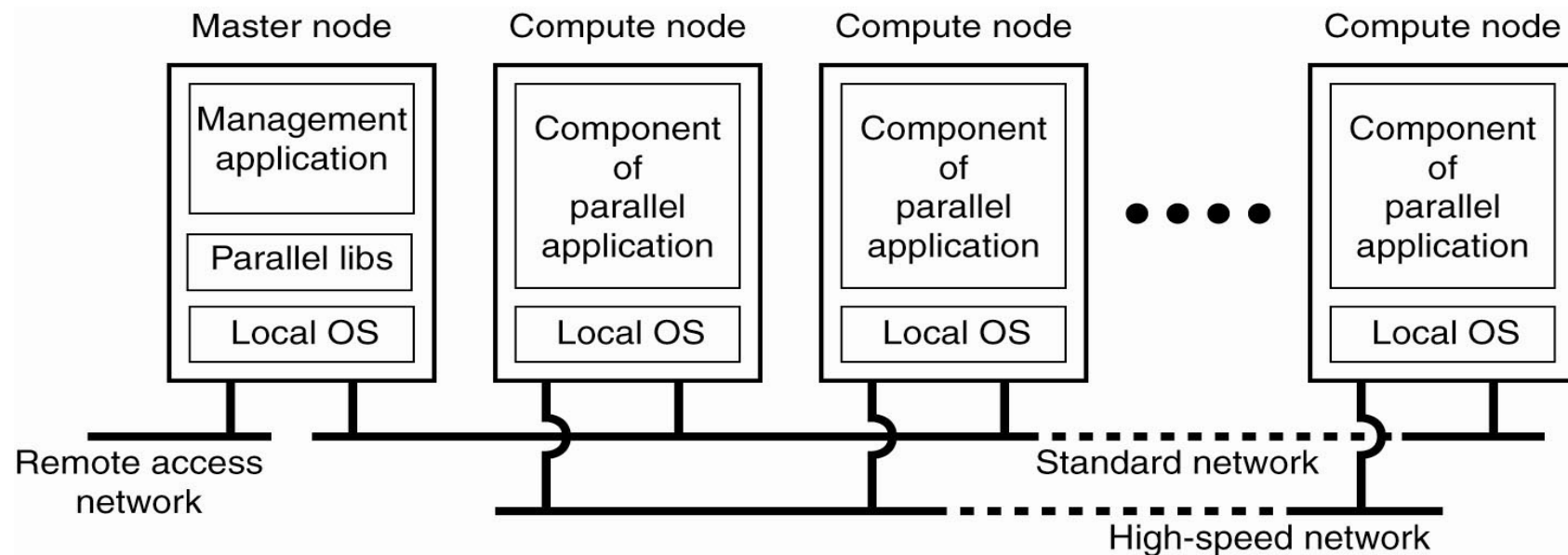
- Underlying hardware consists of a collection of similar workstations.
- Connected by high-speed local-area network.

Grid Computing:

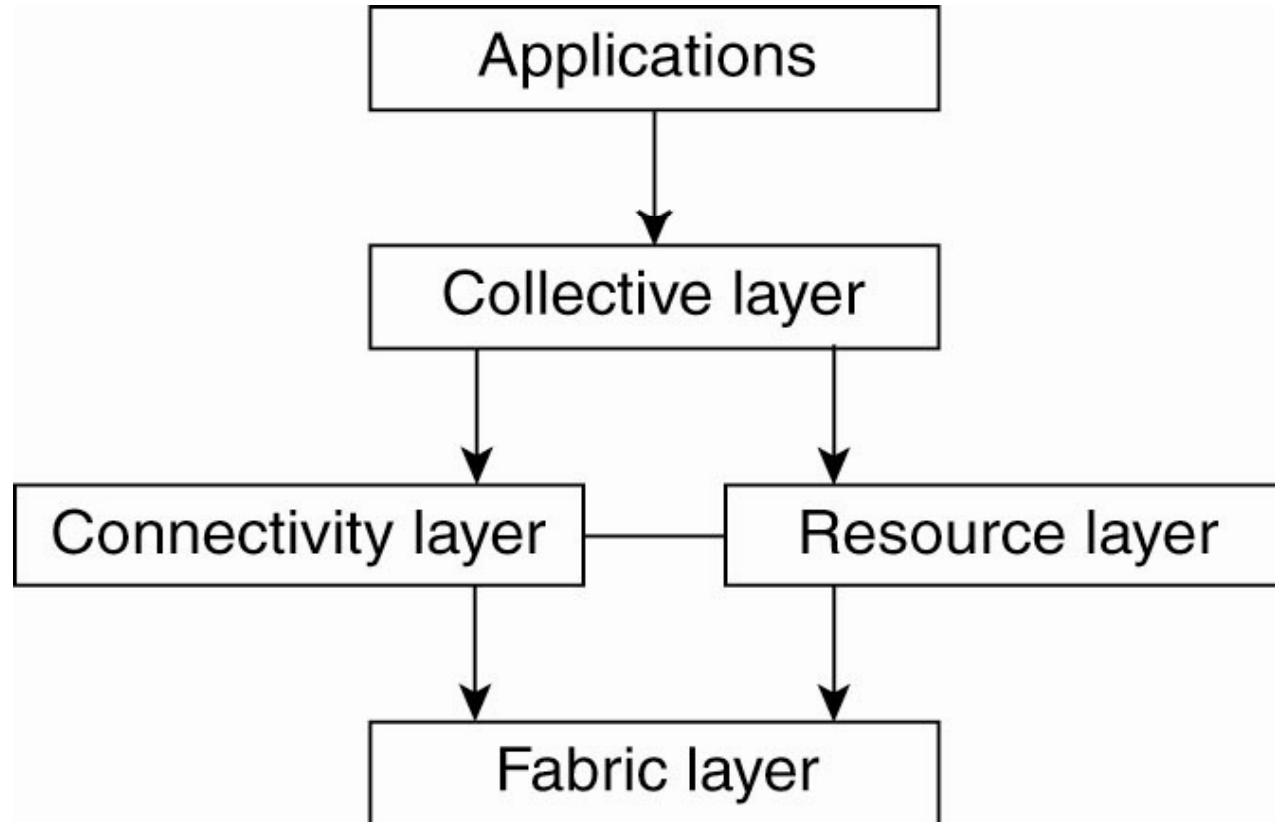
- Different administrative domain.
- Different hardware, software, deployed network technology.

Cluster Computing Systems

- Figure 1-6. An example of a cluster computing system.



Grid Computing Systems



- Figure 1-7. A layered architecture for grid computing systems.

Distributed Information Systems.

Transaction Processing
Systems.

Enterprise Application
Integration.

Transaction Processing Systems (1)

- Figure 1-8. Example primitives for transactions.

Primitive	Description
BEGIN_TRANSACTION	Mark the start of a transaction
END_TRANSACTION	Terminate the transaction and try to commit
ABORT_TRANSACTION	Kill the transaction and restore the old values
READ	Read data from a file, a table, or otherwise
WRITE	Write data to a file, a table, or otherwise